

Weak-Instrument Inference and Residualized Concentration in Double Machine Learning for the Partially Linear Instrumental Variables Model*

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Abstract

We prove that in the partially linear instrumental-variables model, under a correctly specified treatment nuisance, prediction-optimal instrument residualization is concentration-optimal in population: flexible learning imposes no absorption cost. Absent a strong residualized first stage, an estimate collapsing toward zero with a small standard error signals lost residualized concentration, not a precise zero. We make weak-instrument-robust inference the default reporting layer, importing an i.i.d. uniform-size baseline and proving the cross-fitted Anderson–Rubin set uniformly valid under clustering, survey weights, and a fold-nested generated instrument. A four-design diagnostic portfolio, a known-truth Monte Carlo, and an exploratory selector trace realized losses to diagnosable sources.

Keywords: double machine learning, partially linear instrumental-variables model, weak instruments, residualized concentration, identification-robust inference, identification-aware nuisance selection.

*This paper uses the partially linear IV setting and the CollegeDistance application of Çetin (2026a). All numerical results are reproduced by the replication materials accompanying the paper.

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1. Introduction

Double machine learning (DML) lets applied researchers estimate a partially linear instrumental-variables (PLIV) model whose rich controls enter through flexible, high-dimensional learners, while retaining valid $\sqrt{n}$ -inference (Chernozhukov et al. 2018). The flexibility can impose a less obvious cost: under a flexible learner for the instrument-residualization nuisance $\pi_0(W) = \mathbb{E}[Z | W]$, the residualized instrument can become weak. Çetin (2026a), which establishes $\sqrt{n}$ -asymptotic normality for this model under strong identification, documents the sensitivity in its CollegeDistance return-to-schooling application. Regularized-linear residualization leaves the estimate near its two-stage least-squares value. Random-forest nuisance learners (the π -learner among them) drive it to approximately zero with a small Wald standard error. A near-zero estimate beside a small standard error invites the reading of a precise zero, a schooling return that is absent rather than undetermined.

We argue this reading is wrong: absent a strong residualized first stage, a collapse toward zero with a small standard error signals lost residualized concentration, not a precise zero (Proposition 2). Under a correctly specified treatment nuisance, flexible learning carries no population concentration cost (Proposition 3). The pattern first received a different reading.

Çetin (2026a) proposes the mechanism reading, an overly flexible instrument residualization absorbing a large share of the instrument’s identifying variation (its Section 5.1); it shows the small standard errors are not a scaling artifact and co-occur with diagnostics of limited residualized instrument variation (its Table 2); and it names the residualized concentration quantity that governs precision (its Remark 3), while maintaining strong identification throughout. It flags weak-instrument-robust inference as the needed remedy but places it out of scope, reporting Wald standard errors. Two problems remain. First, where the

residualized concentration is low, the Wald interval is not a valid precision statement, so the collapse is reported with a tool the same analysis invalidates. Second, every instrument in the schooling application is conditionally near-irrelevant, so a π -estimation artifact and conditional irrelevance are observationally equivalent there.

The inference we adopt comes from the weak-instrument literature: the Anderson and Rubin (1949) statistic and its refinements (Kleibergen 2005; Moreira 2003), whose confidence sets control size regardless of instrument strength, and a robust first-stage Wald statistic, the Montiel Olea and Pflueger (2013) effective F in the single-instrument case, which screens realized strength.¹ This machinery was developed for fixed instruments in classical two-stage least squares; its behavior on a cross-fitted, generated instrument is not established. We execute the Anderson–Rubin set and the effective- F screen on the cross-fitted PLIV moment and do not assume the classical size control transfers (§3.2 states the theory).

We make five contributions.

- **Weak-instrument-robust inference, executed.** We make effective- F screening and a cross-fitted Anderson–Rubin set the default reporting layer, so the collapse is reported with an identification-robust interval. Theorem 2, proved in Online Appendix A.4, gives the set uniform asymptotic validity under clustering; the i.i.d. baseline is imported from Çetin (2026b).
- **A structured diagnostic portfolio.** Four empirical applications, which we call cells, vary in residualized first-stage strength. In the randomized Oregon Health Insurance Experiment (OHIE) cell the effective- F is about 2100, flat across learners, with no material overfit dilution. In the high-dimensional examiner-leniency cell (Chyn, Frandsen, and Leslie 2025) a random-forest π -learner overfits and dilutes the leniency instrument, cutting the effective- F from the ridge learner’s 20.1 to 14.2 and widening the Anderson–Rubin set. In the CollegeDistance cell the apparent absorption was leakage across geographic folds; with the leakage removed it is simply weak, its Anderson–

¹See also Lee, McCrary, Moreira, and Porter (2022) and Andrews, Stock, and Sun (2019).

Rubin set spanning both signs. In the Card (1995) cell the schooling return is stable across learners; on a fixed subsample, richer controls lower the effective- F from about 7.5 to about 2.5, the conditioning-set route through $r(W)$. The contrast separates estimator pathology from a design already weak after conditioning. The over-identified Angrist–Krueger exercise is reported alongside (Section 4).

- **The concentration view formalized, with a new optimality result.** Under a correctly specified treatment nuisance the prediction-optimal π_0 *maximizes* the population residualized concentration (Proposition 3), so flexibility per se carries no absorption cost. Two framing results follow: the estimand is invariant to the choice of consistent instrument residualization (Proposition 1), and the inferential failure is distributional rather than one of standard-error scale (Proposition 2), so a near-zero estimate with a small standard error is never read as a precise zero (Section 2).
- **A known-truth Monte Carlo in the weak regime.** In the §4.2 evaluation the cross-fitted set keeps coverage close to nominal, and closer than Wald, where identification is weak; the two off-nominal settings are reported as such, Wald the closer at the flexible-RF point. Varying the treatment-nuisance specification separates the misspecified- r interaction from any genuine population cost and finds none (Proposition 3), as observational cells alone cannot (§4.8).
- **An identification-aware π -selector.** Following the identification-aware cross-validation of Çetin (2026b), we screen π -learners by out-of-fold predictive loss and break ties toward residualized strength, both on a selection subsample disjoint from the inference fold. In a paired known-truth benchmark the measured accuracy differences are within Monte-Carlo uncertainty, and we present the selector as an exploratory refinement rather than a demonstrated improvement (Section 4). Its precursor is the restricted- π analysis of Çetin (2026a), a binary library switch; ours is a continuous, rate-condition-constrained tie-break.

Absent a strong residualized first stage, a near-zero estimate under flexible learning is

a weak-instrument, finite-sample, and nuisance-specification phenomenon, and standard diagnostics support attributing a realized concentration loss to its source (§2.4).

The reporting layer travels to other DML-IV applications as four steps. (i) The outcome, treatment, and instrument-residualization nuisances are cross-fitted under the leakage-safe nested protocol, with any learner selection made inside the outer training set (§2.1; Online Appendix F). (ii) The residualized effective- F screens realized strength (§3.2). (iii) The cross-fitted Anderson–Rubin set, valid in the uniform asymptotic sense of §3.2 wherever one of its assumption packages holds, is the primary interval and the only one to interpret where the screen reads weak (§4.1). (iv) The out-of-fold predictive R^2 of the π -learner and the ratio of residualized to raw instrument variance separate overfit dilution from the treatment-nuisance and conditioning route (§2.4).

1.1 Related literature

Our inferential layer builds on Çetin (2026b). Under weak identification, adaptive nuisance learning is itself an inferential act: tuning that reuses inference observations can change the null law of otherwise robust statistics at first order, and prediction-optimal tuning can attenuate residualized identifying strength. That paper’s leakage-free nested cross-fitting (LF-NCF) protocol and identification-aware cross-validation (IACV) restore the required filtration: no selection statistic touches the inference fold. Theorem 3 of that paper, imported here as Theorem 1 (Online Appendix A.4), gives the cross-fitted Anderson–Rubin test uniform asymptotic size over classes including local-to-zero identification. Our designs violate the Theorem 1 baseline’s i.i.d., observed-instrument, and unweighted-target-law assumptions. This paper supplies the four extensions they require (Online Appendix A.4–A.6): clustered sampling with cluster-level fold assignment (Theorem 2), a Kleibergen (2005)-type score statistic (Theorem 3), survey-weighted target laws (Proposition 4), and a generated leave-one-cluster-out instrument (Proposition 5). The Anderson–Rubin results are general; the generated-instrument score claim is limited to the single-instrument case $d_z = 1$, where score

and AR coincide. Çetin (2026c) addresses the complementary power question: when a scalar Anderson–Rubin direction is learned on a training split from a growing instrument dictionary, sample splitting preserves size, but a sparse minimax frontier caps the oracle power any such rule can recover. This paper learns no direction: the PLIV orthogonal moment is fixed, only nuisances are selected, and the inferential results concern size and coverage. Inherited from Çetin (2026a) are the nested nuisance-selection idea and product-rate architecture, the residualized concentration object, the CollegeDistance sensitivity and its mechanism reading, the restricted- π analysis, and the Angrist–Krueger stress idea. We also inherit the ridge-AR and conditional-QLR theorems of Çetin (2026b).

Ma (2026) is the closest published work: an orthogonalized Anderson–Rubin test for the LATE with high-dimensional covariates, with an appendix extension (her Appendix B, Example 1) that covers scores linear in the parameter, the partially linear IV model among them, through a cross-fitted conditional QLR test. Her results share our cross-fitted, Neyman-orthogonal construction and our uniformity ambitions, but they are proved for i.i.d. data, an observed instrument, an unweighted target law, and a nuisance learner fixed in advance subject to a rate condition. Adaptive selection over a learner library is outside her scope, and Çetin (2026b) shows that regime can break pivotality. Clustering, weighting, and generated instruments are outside her scope as well. Relative to Ma, our paper (i) inherits selection-robustness from the LF-NCF filtration rather than fixing the learner ex ante, and (ii) supplies the sampling-design extensions that applied IV work requires.

A second strand is robust to the instrument dimension rather than to the nuisance pipeline. Mikusheva and Sun (2022) develop a jackknife Anderson–Rubin test for instrument counts growing proportionally with the sample size (building on Chao and Swanson 2005; Hansen, Hausman, and Newey 2008), Navjeevan (2025) a jackknife score statistic agnostic to instrument dimension, and Lim, Wang, and Zhang (2024) a bootstrap dimension-agnostic Anderson–Rubin construction. All three treat the instruments as fixed, partial controls out by exact linear projection, and assume independent observations. Navjeevan (2025,

p. 4, fn. 1) notes that it remains “unclear how a test could be constructed that allows for high-dimensional controls and incorporates first stage information”, and that margin is what our orthogonalized, cross-fitted residualization addresses; our setting is the complement, a fixed, moderate instrument dimension with high-dimensional or nonparametric confounding absorbed by cross-fitted machine learners.²

On the clustering margin, Wang and Zhang (2024) prove validity of wild bootstrap Anderson–Rubin inference for linear IV with few large clusters, the complement of Theorem 2’s regime of many moderate clusters. The Angrist–Krueger cell sits near that boundary with fifty-one large clusters. That count exceeds its thirty instruments, as their studentized bootstrap Anderson–Rubin test requires (their Theorem 6.1), so their wild-cluster bootstrap is the natural complementary benchmark for that cell, cited here rather than run.

Belloni, Chen, Chernozhukov, and Hansen (2012) is the earliest antecedent of selection-aware IV inference: lasso selects optimal instruments from a high-dimensional dictionary, and a sup-score statistic provides a weak-identification-robust fallback.³ Our framework locates selection in the nuisance-residualization layer, controls it by the LF-NCF filtration, and states uniform results.

The panel strand is young, and its entries are unpublished working papers as of this writing.⁴

The examiner-design literature instruments a treatment with the leniency of a quasi-randomly assigned decision-maker (Dobbie, Goldin, and Yang 2018; Chyn, Frandsen, and Leslie 2025); our high-dimensional cell uses a fold-nested cluster-jackknife leniency instrument

²The “cross-fit” variance estimators of Mikusheva and Sun (2022) are leave-one-out variance proxies within a full-sample projection statistic, not the fold-based sample splitting that licenses machine learning here. Online Appendix D.6 records the Mikusheva–Sun pre-test reading of the quarter-of-birth design.

³Their selection occurs in instrument construction, their sup-score screens raw instruments after low-dimensional linear partialling, their size control is pointwise, and none of the three PLIV nuisances is cross-fitted away.

⁴Baiardi, Clarke, Naghi, and Polselli (2026) build on a first-difference transformation with a Neyman-orthogonal score, assign each unit’s entire time series to one fold, and apply a panel analogue of the first-stage F statistic and an Anderson–Rubin test to shift-share migration designs. Avila Márquez (2026) shows that linear reduced-form specifications can make panel instruments spuriously weak and proposes a Super Learner control-function estimator with a generalized concentration parameter, a route adjacent to, rather than an instance of, the orthogonal-score framework used here.

and adds the dilution-versus-absorption diagnosis and the residualized effective- F accounting (§4.5; Online Appendix D.3).

A parallel flexible-instrumental-variables literature targets heterogeneous or structural objects (Athey, Tibshirani, and Wager 2019; Hartford et al. 2017) or studies the implicit weighting of the linear IV estimand under flexible controls (Blandhol, Bonney, Mogstad, and Torgovitsky 2026); its estimands differ from the scalar partially linear effect, so a like-for-like comparison would first require testing for homogeneity.

Section 2 sets out the PLIV model and identification, Section 3 estimation and inference, Section 4 the applications, Section 5 limitations, and Section 6 concludes. Online Appendices A–C collect the proofs, full result tables, and Monte-Carlo designs; Online Appendices D–F the cell-level, selector, and protocol detail.

2. Model and Identification

2.1 Setup

We study the partially linear instrumental-variables (PLIV) model

$$Y = \tau_0 D + g_0(W) + \zeta, \quad \mathbb{E}[\zeta \mid Z, W] = 0,$$

where D is a scalar endogenous regressor, Z is an instrument of dimension d_z , W is a vector of controls, and τ_0 is the scalar structural parameter of interest. Estimation rests on the residualized, Neyman-orthogonal PLIV score

$$\psi(\tau, \eta) = \left(Z - \pi(W) \right) \left[(Y - \mu(W)) - \tau (D - r(W)) \right],$$

where $\eta = (\mu, r, \pi)$ collects the nuisances, whose true values (subscript 0) are $\eta_0 = (\mu_0, r_0, \pi_0)$ with $\mu_0(W) = \mathbb{E}[Y \mid W]$, $r_0(W) = \mathbb{E}[D \mid W]$, and $\pi_0(W) = \mathbb{E}[Z \mid W]$. We write residualized variables as $\tilde{Y} = Y - \mu_0(W)$, $\tilde{D} = D - r_0(W)$, and $\tilde{Z} = Z - \pi_0(W)$. A naming caution: the

classical “first stage” regresses the treatment on the instruments, and π_0 is not that object. We call π_0 the **instrument-residualization nuisance** (the residualization, for short). The classical first stage appears here only in residualized form, as the $\tilde{D}$ -on- $\tilde{Z}$ relation whose strength §2.3 and §3.2 measure.

We take as given the leakage-safe nested cross-fitting of Çetin (2026a,b), run in the three-way form of Çetin (2026b, Algorithm 1). Nesting matters because, under weak identification, selection that reuses inference observations can distort the null law of identification-robust statistics at first order (Çetin 2026b). The protocol splits the complement of each held-out (inference) fold into an estimation and a selection subsample. Candidate learners for (μ, r, π) are fit on the estimation subsample, selected on the selection subsample, and the selected fits evaluate the held-out fold, with no refit on the union. Every selected object is therefore measurable with respect to data disjoint from the fold it scores. §4.3 equips the π slot with an identification-aware selection rule. Under a product-rate condition and strong identification (the residualized concentration bounded away from zero), Çetin (2026a, Theorem 2) delivers $\sqrt{n}$ -asymptotic normality of the residualized-IV estimator. That theorem is stated for the protocol’s inner-cross-validation form, run in parallel here as a robustness check (Table B.8; Online Appendix F). Our contribution begins where the strong-identification assumption is relaxed.

2.2 Estimand invariance across instrument residualizations

The role of π in the score is to residualize the instrument, not to identify τ_0 .

Proposition 1 (Estimand invariance and the role of the instrument residualization). *Suppose the PLIV model holds with $\mathbb{E}[\zeta \mid Z, W] = 0$, that the outcome and treatment nuisances are consistent ($\hat{\mu} \rightarrow \mu_0$, $\hat{r} \rightarrow r_0$), and that $\mathbb{E}[\tilde{Z}\tilde{D}] \neq 0$. Then for **any** residualization estimator $\hat{\pi}$ converging to a square-integrable, W -measurable limit $\pi_*(W)$ (not necessarily π_0), the cross-fitted residualized-IV estimator is consistent for τ_0 . Under these hypotheses the choice of residualization affects only the residualized concentration C , and through it precision, not*

the estimand or consistency (Online Appendix A.1 records the scope of the precision claim).

Since the residualized instrument is a valid multiplier for *any* residualization (Online Appendix A.1), two consequences follow. First, a collapse of $\hat{\tau}$ toward zero under flexible learning is not by itself evidence that “the true effect is zero” or that “identification was destroyed.” Second, the estimator built on a linear residualization remains consistent for τ_0 *even when π_0 is nonlinear*, because the linear projection is an admissible limit π_* ; when two learners disagree, neither is biased on residualization grounds, and the disagreement is about concentration (or a misspecified outcome or treatment nuisance), not about the instrument. Both consequences are statements *within* the homogeneous-coefficient PLIV model. Under unmodeled effect heterogeneity the estimand of a residualized-IV estimator is a weighted average of heterogeneous responses whose weights depend on the residualized instrument, so a change of residualization can change that pseudo-target; the invariance above is exact only under the maintained homogeneity, and the target map of §4.1 (Table 2) records where each application maintains it.

2.3 Concentration governs precision, and the failure mode is distributional

Define the per-observation residualized concentration

$$C = g'_{ZD} \Omega^{-1} g_{ZD}, \quad g_{ZD} = \mathbb{E}[\tilde{Z}\tilde{D}], \quad \Omega = \mathbb{E}[\tilde{Z}\tilde{Z}'],$$

and the concentration parameter $\mu^2 = nC$. The classical concentration parameter divides nC by the first-stage error variance and is dimensionless; our μ^2 keeps the treatment scale and enters only order and ranking statements within one design. The asymptotic variance of the residualized-IV estimator scales as $\text{Var}(\hat{\tau}) \propto 1/(nC) = 1/\mu^2$ under conditional homoskedasticity of the score; under bounded conditional heteroskedasticity C ranks weak-identification strength up to constants (Online Appendix A.1), so a reduction in C is a loss

of identification strength at fixed n .

The population concentration C is a projection index and the only object Proposition 3 (§2.4) orders: with $r = r_0$, $C(\pi) \leq C(\pi_0)$ across residualizations. The *effective- F* reported in §3–§4 is a sample statistic: the robust first-stage Wald statistic divided by the numerical rank of the residualized instrument covariance (equal to d_z in every reported run), computed on the residualized data; it screens realized strength and is neither C nor a variance. The sandwich variance V measures precision; it is tied to C exactly only under conditional homoskedasticity of the score (Online Appendix A.1). Population claims below are statements about C , realized-strength statements are about the effective- F , and precision statements are about V .

Proposition 2 (Distributional, not scale, failure). *In the just-identified case $d_z = 1$, write $u = \tilde{Y} - \tau_0 \tilde{D}$ for the structural score residual, $\Sigma = \mathbb{E}[\tilde{Z}^2 u^2]$, and $V = \Sigma / g_{ZD}^2$ for the Wald variance. (i) This V is **invariant to the scale of $\tilde{Z}$** : the magnitude of the reported standard error therefore carries no evidence about instrument strength on its own, in either direction. (ii) Under the local-to-zero sequence of Staiger and Stock (1997), $g_{ZD,n} = \gamma / \sqrt{n}$ (equivalently, nC bounded as $n \rightarrow \infty$), the cross-fitted joint limit of Lemma 1 (Online Appendix A.2) yields: $\hat{\tau}$ is inconsistent and $O_p(1)$; the standard error neither vanishes nor explodes ($\hat{\text{se}}$ and $\hat{\text{se}}^{-1}$ are both $O_p(1)$); and the statistic whose variance is evaluated at the hypothesized value is asymptotically pivotal in absolute value, its square being exactly the cross-fitted AR statistic of §3. (iii) The feasible Wald statistic $(\hat{\tau} - \tau_0) / \hat{\text{se}}(\hat{\tau})$, whose variance evaluates residuals at $\hat{\tau}$, is **non-pivotal** along the sequence for every value of the score correlation $\rho = \text{Corr}(\tilde{Z}u, \tilde{Z}\tilde{D})$, including $\rho = 0$, and is standard normal only in the strong-identification limit. Standard-normal critical values are therefore not generically valid along the sequence, no fixed critical value controls size uniformly over strength and score-correlation configurations, and the nominal $1 - \alpha$ Wald interval does not attain $1 - \alpha$ coverage in general (Online Appendix A.2).*

Proposition 2 corrects the tempting reading of the schooling collapse as a precise zero. By part (i), the magnitude of the conventional standard error is not a diagnostic of instrument

strength: holding the score variance fixed, a smaller $|g_{ZD}|$ *raises* the Wald variance. What breaks under weak identification is the shape of the sampling distribution: by part (iii), the studentized statistic is not asymptotically standard normal even when the score correlation vanishes, while the null-normalized statistic of part (ii) has no such defect. Hence (§3) identification-robust inference from inverting the null-normalized statistic must be the primary reporting layer whenever C is low, not an optional diagnostic.

We do not discover this pattern: as credited in §1, Çetin (2026a) documents the sensitivity, shows that the small standard errors under the collapse are not a scaling artifact and co-occur with diagnostics of limited residualized instrument variation, and names the concentration quantity $g'_{ZD}\Omega^{-1}g_{ZD}$, all under maintained strong identification. We add the anatomy: Propositions 1–2 formalize the sensitivity as a concentration-and-distribution statement, and Proposition 3 shows that its natural reading (flexible π *absorbing* the instrument) does not survive under correctly specified nuisances (§2.4).

2.4 The anatomy of concentration loss

Proposition 2 says low C breaks Wald inference, not what lowers it.

Proposition 3 (Concentration is maximized at the true π_0). *Assume $\mathbb{E}\|Z\|^2 + \mathbb{E}[D^2] < \infty$, that $\Omega(\pi_0)$ is positive definite, and that the relevance vector $g = \mathbb{E}[\tilde{Z}_0\tilde{D}]$ is nonzero. With the treatment nuisance correctly specified, $r = r_0 = \mathbb{E}[D \mid W]$, the residualized concentration $C(\pi) = g_{ZD}(\pi)' \Omega(\pi)^{-1} g_{ZD}(\pi)$ satisfies $C(\pi) \leq C(\pi_0)$ for every square-integrable W -measurable π , with equality iff $\pi = \pi_0$ when $d_z = 1$, and for $d_z > 1$ iff the perturbation $\pi_0 - \pi$ is concentration-neutral ($M \Omega(\pi_0)^{-1} g_{ZD} = 0$, with $M = \mathbb{E}[(\pi_0 - \pi)(\pi_0 - \pi)']$; Online Appendix A.3). The prediction-optimal residualization π_0 is therefore concentration-optimal in population: a consistent flexible learner that recovers a nonlinear π_0 attains the **maximum** concentration and cannot lower it.*

The proof (Online Appendix A.3) closes only the population channel: realized, finite-sample concentration can still fall, through two diagnosable routes separated by the out-of-fold

predictive R^2 of $\hat{\pi}$ on Z ($R_{\text{of}}^2(\hat{\pi}; Z)$) and the residualized-variance ratio $\text{Var}(\tilde{Z})/\text{Var}(Z)$.

Diagnostic characterization (routes by which the *realized* concentration falls). •

Overfit dilution: a finite-sample π -*estimation* pathology. $\hat{\pi}$ has no out-of-fold predictive gain, $R_{\text{of}}^2(\hat{\pi}; Z) \leq 0$, yet a large in-sample overfit gap; cross-fitting converts the overfit into idiosyncratic noise in the held-out residual, so $\text{Var}(\tilde{Z})/\text{Var}(Z) > 1$ and g_{ZD} is diluted. The dilution vanishes as $n \rightarrow \infty$ and is removed by a restricted or cross-fit-stabilized $\hat{\pi}$.

- **Treatment-nuisance and conditioning effects:** operating through r and W , not π :
 - (i) a richer set of controls W can lower the *conditional* relevance $\text{Cov}(\tilde{Z}, \tilde{D})$; a modeling choice, read off C (the Card ladder, §4.7); (ii) an *under-specified* $\hat{r} \neq \mathbb{E}[D | W]$ breaks the π -invariance of g_{ZD} that underlies Proposition 3 and can manufacture an apparent instrument “absorption,” which a correctly specified flexible $\hat{r}$ removes (§4.8). Under stratified randomization of Z given W the assignment shares are known within strata, so π_0 equals those shares; a learner can at most recover them, residualization removes only the across-strata variance of Z , and C is preserved in population up to that design variation. In-fold noise fitting remains possible in principle; it is what the dilution diagnostics screen, and §4.4 shows it is not materially present in the randomized cell.

$R_{\text{of}}^2(\hat{\pi}; Z) > 0$ with $\text{Var}(\tilde{Z})/\text{Var}(Z) < 1$ means the learner is recovering a predictable component of π_0 (the linear baseline judges whether it is nonlinear); under correct nuisances this *raises* C (Proposition 3), so this signature alone is not a loss. A loss alongside it admits a misspecified- r interaction, a change in conditioning, or finite-sample noise; the diagnostics separate these only jointly. The examiner cell instantiates overfit dilution, the Card ladder the conditioning/ r -route, and the known-truth Monte Carlo the misspecified- r interaction; the OHIE cell is the randomized control where neither route is materially present. No cell exhibits a population absorption cost of flexible π , and by Proposition 3 none can.

3. Estimation and Inference

3.1 Cross-fitted residualized estimation

We cast each dataset into (Y, D, Z, W) , estimate $(\hat{\mu}, \hat{r}, \hat{\pi})$ by the leakage-safe three-way protocol of §2.1 with cluster-aware folds, form the held-out residuals $(\tilde{Y}, \tilde{D}, \tilde{Z})$, and solve the empirical analogue of $\mathbb{E}[\psi] = 0$. For $d_z > 1$ we use the Ω^{-1} -weighted (two-stage-least-squares-style) residualized GMM estimator

$$\hat{\tau} = \frac{\hat{g}'_{ZD} \hat{\Omega}^{-1} \hat{g}_{ZY}}{\hat{g}'_{ZD} \hat{\Omega}^{-1} \hat{g}_{ZD}}, \quad \hat{g}_{ZY} = \frac{1}{n} \sum_i \tilde{Z}_i \tilde{Y}_i,$$

with the standard sandwich variance. Cross-fitting folds are **cluster-aware**: whole clusters (judge shifts, households) are assigned to a single fold, so within-cluster information cannot leak across the train/held-out split. First-stage F statistics and identification-robust statistics carry each cell’s cluster structure (households, judge shifts, the CollegeDistance state proxy, state of birth in the Angrist–Krueger cell); the Card cell maintains no clusters, so its statistics are heteroskedasticity-robust. Wald standard errors follow the same levels, except that the CollegeDistance OLS and 2SLS rows are reported with heteroskedasticity-robust standard errors; the per-cell levels are recorded in the Table 1 notes and in Table B.1; for that cell the notes also report the conventional 2SLS clustered standard error and robust interval under the same grouping. The outcome, treatment, and instrument-residualization nuisances are estimated with **separate** learners, each selected by its own out-of-fold prediction loss; this isolates the π -learner under study and prevents a misspecified outcome or treatment nuisance from being confounded with it (the coupling behind the spurious learner disagreement of §4.6). Hyperparameter tuning is delegated to each candidate learner’s internal cross-validation on its own training data, so the outer nested selection chooses among learner families only and every tuned value remains measurable with respect to the inference fold’s training complement. In the decoupled cells the instrument-residualization learner is held fixed rather than selected;

the fixed $\hat{\pi}$, involving no selection step, is cross-fit on the full $(K - 1)$ -fold complement of each inference fold, uniformly across cells and selection protocols.

3.2 Identification-robust inference is the primary layer

Cetin (2026a) flags weak-instrument-robust inference as needed but out of scope (Remark 3 and Conclusion) and reports Wald standard errors. By Proposition 2, in any low-concentration cell those Wald intervals are not valid precision statements. We therefore execute identification-robust inference as the default reporting layer, not a supplementary diagnostic.

We first **screen** with the effective- F of §2.3, computed on the cross-fitted residualized moment. We read it against the worst-case-bias tables of Montiel Olea and Pflueger (2013) rather than the rule of ten, noting two limits: the implemented statistic coincides with their effective F only at $d_z = 1$, and their critical values are derived for 2SLS and LIML in the linear IV model, not for cross-fitted DML statistics, so at $d_z > 1$ the tables serve only as a heuristic screen.⁵ We therefore rely on the identification-robust sets, not the effective- F threshold, to assess inference (Theorems 1 and 2 below; coverage evaluated in §4.2). We report, for every estimator row, a cross-fitted **Anderson–Rubin (AR)** confidence set, inverted analytically at $d_z = 1$ and over an adaptive grid at $d_z > 1$ (Online Appendix A.4 records the inversion mechanics), alongside the Wald interval; we flag rows where the two diverge. For $d_z = 1$ (every empirical cell except the over-identified Angrist–Krueger design of §4.9) the AR, score, and conditional-QLR sets coincide, so we report the AR set directly. In an over-identified design, the Kleibergen (2005) score (LM) statistic directs power at the τ direction; the conditional QLR statistic adapts between the score and the full-vector AR (Online Appendix A.5); and over-identification is tested by $J_{\text{CUE}} = \min_{\tau} \text{AR}(\tau)$, the continuously-updated GMM objective (Stock and Wright 2000), computed as a grid minimum (Online Appendix A.5 bounds the numerical error), which plays the role of the classical Sargan–Hansen test (Sargan 1958; Hansen 1982) and is read against χ^2_{r-1} , its strong-identification reference; here

⁵With multiple endogenous regressors, a case that does not arise here, the first-stage screen would instead be the conditional F of Sanderson and Windmeijer (2016).

r , the numerical rank of the estimated score covariance at the minimizer, equals d_z in every reported run. Because $\min_{\tau} \text{AR}(\tau) \leq \text{AR}(\tau_0)$, and Theorem 2 gives $\text{AR}(\tau_0)$ a uniformly valid $\chi^2_{d_z}$ reference, reading J_{CUE} against χ^2_r critical values is size-valid, conservatively, under weak identification; this conservative p -value is reported alongside the statistic (Table B.6). The Angrist–Krueger quarter-of-birth cell (§4.9) exercises all three, with cluster-robust meats. The clustered score statistic retains its uniform χ^2_1 size with the cluster-summed meats (Theorem 3, Online Appendix A.5). The clustered QLR variant plugs the cluster meats into the same construction and carries no distributional claim here; the §4.2 simulations evaluate its i.i.d. analogue at the Angrist–Krueger instrument counts and strengths. The AR variance is computed with a heteroskedasticity- or cluster-robust per- τ form, because the residualized moment carries generated-regressor heteroskedasticity, and the set is reported as a set, including unbounded or empty outcomes, following Davidson and MacKinnon (2014).

The PLIV moment $\mathbb{E}[\tilde{Z}(\tilde{Y} - \tau\tilde{D})]$ is first-order Neyman-orthogonal in all three nuisance directions at *every* τ (Online Appendix A.4 records the reason). What survives is second order: the products of the $\hat{\pi}$ estimation error with the outcome- and treatment- nuisance errors enter the moment, the treatment product scaled by the tested τ . The product-rate condition controls these terms at any fixed τ , but an inverted confidence set requires that control at every candidate τ , including remote values where the τ -scaled product term is largest, and the generated instrument contributes estimation noise of its own. A $\chi^2_{d_z}$ reference for the cross-fitted AR statistic is therefore not automatic: uniform validity is exactly what Theorems 1 and 2 establish. The §4.2 Monte-Carlo evaluation complements these asymptotic statements at each study’s realized dimensions and strengths.

Theorem 1, imported from Çetin (2026b, Theorem 3), is the i.i.d. baseline: under its Assumptions 1–7 (A1–A7 here), the cross-fitted AR statistic at the tested value has uniform asymptotic size α against the $\chi^2_{d_z}$ reference, and the inverted set has uniform asymptotic coverage $1 - \alpha$. Online Appendix A.4 maps the implemented §2.1 protocol to A1–A7. Theorem 2, this paper’s clustered extension, delivers the same uniform-size and uniform-

coverage conclusions as $G \rightarrow \infty$ with d_z fixed, over classes containing cluster-level $G^{-1/2}$ relevance drifts, under conditions C1–C8 (proved in full in Online Appendix A.4). Neither theorem imposes a relevance condition: the score is mean zero at τ_0 whatever the instrument’s strength.

Theorem 3, also this paper’s, treats the cross-fitted score statistic: under A1–A7 (i.i.d.) or C1–C8 (clustered), together with conditions K1 and K3 (Online Appendix A.5), it has uniform asymptotic size at the tested value against χ_1^2 ; at $d_z = 1$ it coincides with the AR statistic, so the theorem adds content only for the over-identified cell. Both the uniformity and the clustered statement go beyond Kleibergen (2005) (Online Appendix A.5, closing remarks).

Proposition 4 (Online Appendix A.6) transfers all three conclusions to the survey-weighted statistics under bounded weights, a weighted-score non-degeneracy condition, and the primitive per-tested-value null W-ID(t), which presumes no solved weighted estimand under weak identification. When the weights are a function of the included controls (regime (a)), unweighted nuisance estimation is valid; for general weights (regime (b)), every nuisance must be estimated and selected under the weighted criterion, with rates in $L^2(P_\omega)$. Proposition 5 (Online Appendix A.6) covers the examiner design’s constructed leniency instrument: fold-nesting the construction makes it a fourth cross-fitted nuisance, and under C1–C8, the exclusion condition G-EX, a leniency-map rate, and the augmented product rate, Theorem 2 holds for the resulting statistic; the score statistic is covered at $d_z = 1$, where it coincides with the AR statistic, and no transfer of Theorem 3 is claimed at $d_z > 1$. Fold-nesting is a hypothesis of the result, not a refinement; the leniency rate asks the judge dimension to grow slowly ($J_n = o(\sqrt{n})$); and no lower bound on the generated instrument’s relevance is imposed.

Two implementation corners sit outside these results and stay flagged. The numerical rank guard’s χ_r^2 convention, when the guard flags rank $r < d_z$, is a diagnostic layer, not a claim covered by the theorems (Online Appendix A.4). And Theorem 2 is a $G \rightarrow \infty$ statement:

the examiner and OHIE cells sit squarely inside its regime, while the Angrist–Krueger cell’s $G = 51$ state-of-birth clusters sit at its boundary, so we do not read the theorem as finite- G evidence there (§4.9).

4. Empirical Applications

4.1 Design: a structured diagnostic portfolio in residualized strength

A portfolio of weak instruments cannot separate an estimator pathology from a design in which the instrument is already conditionally irrelevant, because “flexible $\hat{\pi}$ absorbs Z ” and “ Z was conditionally irrelevant” are then observationally equivalent. We instead treat the residualized first-stage strength as the object of interest and assemble a **structured diagnostic portfolio** of four cells: three chosen so that strength spans its support (a randomized cell where π_0 is known by design, a high-dimensional cell where π_0 is a flexible function of W , and a weak cell where the instrument is known to be weak after conditioning), and the Card cell (§4.7), which supplies the clean schooling estimand and a within-instrument concentration ladder. The portfolio is a set of structured diagnostics, not a designed experiment in strength: population, outcome, treatment, instrument, controls, and dependence all change across cells, and Table 2 states each cell’s statistical and causal target. The weak cell (CollegeDistance, §4.6) doubles as a stress test of the cross-fitting protocol, a leakage-and-misspecification illustration rather than a clean mechanism cell. The cross-cell synthesis is Figure 1, displayed after Table 2 and read in full in §4.8. The over-identified Angrist–Krueger cell (§4.9) is reported separately: it tests the over-identified inference of §3.2 and §4.2 rather than the portfolio contrast.

The cells span the identification spectrum. The Oregon lottery is the strong-identification case (effective- $F \approx 2100$): the Wald and Anderson–Rubin intervals coincide, and the point estimate carries its usual interpretation. Card’s lean specification is moderate but identified (effective- F 15–18); the examiner cell sits between regimes, moderate under a ridge π (20.1) but pushed weak by flexible-learner dilution (14.2). CollegeDistance under its flexible learners

(7.9–8.9), the upper rungs of the Card control ladder (1.9–3.5), and the $d_z = 30$ Angrist–Krueger block are the weak-identification cases: there, following §3.2, the Anderson–Rubin set is the only interval we interpret, and a near-zero estimate beside a small Wald standard error is read under Proposition 2, not as a precise zero. In the strong-identification cells the cluster-aware discipline merely confirms the design; in the weak cells it reverses the CollegeDistance diagnosis (§4.6).

Descriptive statistics and covariate balance are reported in Table D.1 in the Online Appendix, and Table D.2 there reports the residualized first-stage strength by nuisance learner. Table 1 reports the main estimates with their identification-robust intervals. Table 2 then states, for each cell, the statistical and causal target and the evidence layer the cell’s inference rests on, naming the covering formal result of §3.2 and the matching §4.2 simulation arm. The (quasi-)randomization the OHIE and examiner designs rely on is established in the source studies rather than recomputed here.

Table 1: Main results: estimates and identification-robust inference

Cell	Estimator	$\hat{\tau}$	Wald SE	eff.- F	AR 95% set	N
OHIE	OLS	1.196	0.063	—	—	23,441
	2SLS	1.034	0.173	2098	—	23,441
	DML (ridge)	1.037	0.174	2093	[0.70, 1.38]	23,441
	DML (RF)	1.050	0.175	2024	[0.71, 1.39]	23,441
	DML (GB)	1.041	0.172	2094	[0.70, 1.38]	23,441
Examiner	OLS	-0.158	0.005	—	—	91,282
	2SLS	-0.497	0.189	24.4	—	91,282
	DML (ridge)	-0.480	0.223	20.1	[-1.02, -0.04]	91,282
	DML (RF)	-0.439	0.263	14.2	[-1.12, 0.10]	91,282
	DML (GB)	-0.455	0.219	20.6	[-0.98, -0.02]	91,282
CollegeDistance	OLS	-0.001	0.001	—	—	4,739
	2SLS	0.052	0.020	9.6	—	4,739
	DML (ridge)	-0.003	0.113	14.8	[-0.227, 0.766]	4,739
	DML (RF)	-0.006	0.106	7.9	[-0.263, 0.346]	4,739
	DML (GB)	0.001	0.118	8.9	[-0.287, 0.381]	4,739
Card	OLS	0.075	0.004	—	—	3,010
	2SLS	0.132	0.054	13.5	—	3,010
	DML (ridge)	0.127	0.052	15.9	[0.03, 0.27]	3,010
	DML (RF)	0.152	0.054	15.2	[0.05, 0.31]	3,010
	DML (GB)	0.118	0.049	18.2	[0.02, 0.24]	3,010

Notes. $\hat{\tau}$ with cluster-robust (OHIE on household, examiner on the court-by-date bond-hearing shift, CollegeDistance DML rows on the state proxy) or heteroskedasticity-robust (Card, and the CollegeDistance OLS and 2SLS rows) Wald standard errors. The effective- F is the weak-instrument screen on the cross-fitted residualized first stage; at $d_z = 1$, as in every cell of this table, it equals the Montiel Olea–Pflueger effective F (the shared cross-fitted screen for every row, 2SLS included; the CollegeDistance 2SLS row’s own conventional first-stage F closes these notes); the **AR 95% set** is the cross-fitted Anderson–Rubin interval, the primary weak-instrument-robust interval (§3.2), and is the column to read when the effective- F is small. The DML (ridge) row uses a genuinely cross-validated ridge $\hat{\pi}$ (λ selected over eight decades); it coincides with the unpenalized linear benchmark on every cell (effective- F within 0.2 at the moderate-strength cells and within 0.03% at OHIE’s ≈ 2100 ; point estimates within 0.006), so separate linear rows are omitted as redundant. Appendix B (Table B.1) carries the per-fold selection winners, clustering and weighting fields, and per-learner instrument-residualization diagnostics for every cell, with an em-dash marking any field a cell’s artifact does not store. The over-identified Angrist–Krueger cell is reported separately in §4.9. Reading across cells: where the residualized first stage is strong (OHIE) or moderate but real (Card) the estimate is stable across learners with a bounded AR set; in the weak conditional design (CollegeDistance) the DML estimate sits near zero with a wide AR set that spans both signs, so the cell determines neither sign nor magnitude, and the seemingly significant 2SLS estimate is not corroborated by the identification-robust interval. On the same sample, controls, and 43-group clustering as its DML rows, conventional CollegeDistance 2SLS has cluster-robust SE 0.0646, conventional AR 95% interval [-0.096, 0.234], and first-stage F 11.75: the clustering convention alone removes the significance.

Table 2: Statistical and causal targets by application

Cell	Estimand and moment	Causal assumptions (source)	Target population and weights	Dependence; d_z	Evidence layer
OHIE	weighted PLIV coefficient τ_w on ever-Medicaid (binary D); survey-weighted orthogonal score	lottery randomization within household-size strata; exclusion: selection affects visits only through coverage (Finkelstein et al. 2012)	survey-weighted twelve-month responders (weights re-weight toward the eligible population); responder selection bounded by Lee trimming (Table B.5)	household clusters; $d_z = 1$	Prop. 4 (regime (b)) with Thm. 2 (household clusters); clustered and survey-weighted §4.2 arms
Examiner	PLIV coefficient on pretrial release (binary D); generated fold-nested cluster-jackknife leniency instrument	quasi-random assignment within court-by-time, balance per Chyn, Frandsen, and Leslie (2025); exclusion through release	Miami-Dade cases of judges with at least 200 cases (source-study restriction)	court-by-date shift clusters; $d_z = 1$ (generated)	Thm. 2 via Prop. 5 (fold-nested construction); combined §4.2 arm (Table B.2)
College-Distance	diagnostic stress test, not a clean causal cell; PLIV coefficient on years of education (multivalued D)	conditional-on-observables only; region-determined outcome undermines a clean exclusion reading (§4.6)	complete cases with positive wage; folds on the 43-group (region, tuition) state proxy	state-proxy groups; $d_z = 1$	Thm. 2 ($d_z = 1$, 43 state-proxy groups; modest G); clustered §4.2 arm; §4.6 leakage diagnostics
Card	PLIV coefficient on years of education (multivalued D)	conditional independence of college proximity given controls; maintained, not testable at $d_z = 1$	full NLS young-men complete-case sample ($N = 3,010$); ladder on the 2,182-man background-complete subsample	heterosked.-robust; $d_z = 1$	Thm. 1 (imported; Çetin 2026b); i.i.d. §4.2 arms
Angrist-Krueger	PLIV coefficient on years of schooling (multivalued D)	quarter-of-birth exclusion and independence as maintained in Angrist and Krueger (1991); many-weak caveats of Bound, Jaeger, and Baker (1995)	1980 Census men born 1930–39	state-of-birth clusters (51); $d_z = 3, 30$	Thm. 2 (AR) and Thm. 3 (score; K1 non-degeneracy), via the within-transformation bridge Lem. 9, with the $G = 51$ boundary caveat (§4.9); clustered QLR reported as a construction; i.i.d. over-identified §4.2 arms at matched counts and strengths

Notes. Every cell maintains the homogeneous-coefficient PLIV approximation of §2.1; that maintenance is what fixes the estimand across residualizations (§2.2). Where D is multivalued (years of education or schooling), the estimand under effect heterogeneity would be an average-causal-response-type weighted average of per-unit responses, not a binary local average treatment effect; where D is binary (OHIE, examiner), a complier-average reading would additionally require the usual monotonicity condition, which we do not impose. Our estimand is the PLIV coefficient throughout. The evidence layer names the formal result of §3.2 (Appendix A.4–A.6) that covers the cell’s configuration, together with the §4.2 Monte-Carlo arm whose dependence and instrument structure match it. The arms evaluate finite-sample coverage at each study’s realized instrument dimensions and strengths, under the stylized cluster structures documented in Appendix C; the clustered conditional-QLR variant carries no distributional claim, its i.i.d. analogue being evaluated in §4.2.

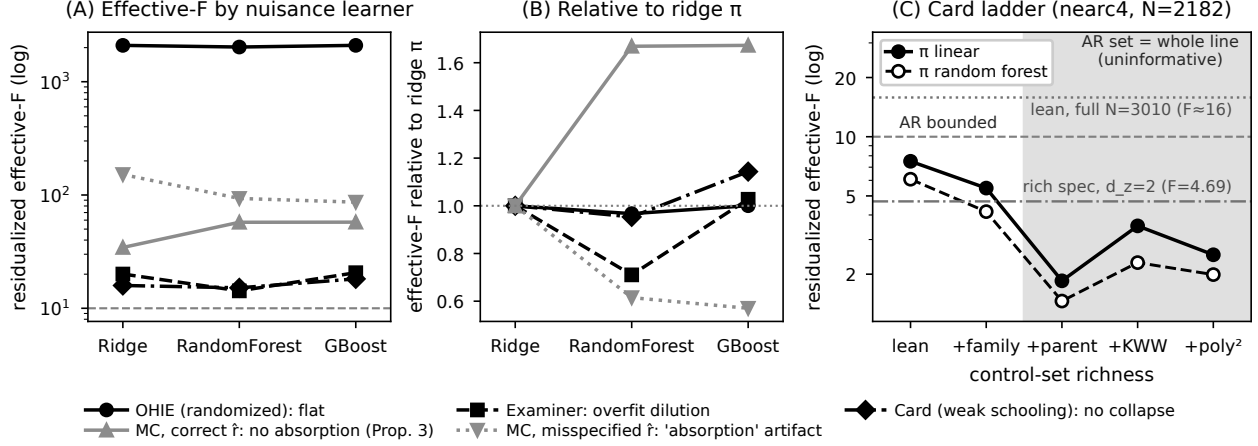


Figure 1: Cross-cell synthesis. (A) residualized effective- F by nuisance learner across cells and the Monte-Carlo two-specification contrast; (B) the same relative to the ridge- π baseline (> 1 gain, < 1 loss); (C) the Card control-richness ladder (instrument fixed) along which richer controls lower concentration through the treatment channel $r(W)$. Correctly specified flexible learning never costs concentration in population (Proposition 3); realized losses are overfit dilution (examiner), conditioning/ r effects (Card), or an under-specified treatment nuisance (Monte Carlo).

4.2 A Monte-Carlo coverage evaluation for the cross-fitted AR set

Table 1 takes the cross-fitted Anderson–Rubin set as the primary interval. A known-truth Monte Carlo shows this set keeps coverage close to nominal across the regimes the empirical cells occupy and, apart from the stress corners reported below, closer to nominal than Wald in the weak regimes; Monte-Carlo standard errors accompany every coverage estimate (Table B.2 prints the Wald and naive variants), and Table B.9 adds what the reported set does: its shape, its length, and whether extracting it recovers the verdict of testing at the truth. The data-generating process is the PLIV model with known (τ_0, π_0) , swept in realized effective- F from about 1 to about 500, with $d_z \in \{1, 3, 5, 10, 30\}$ and both oracle and cross-fitted nuisances. The primary sweep adapts the Monte-Carlo design of Çetin (2026a) to the weak-identification regime; the other six arms below are new. Online Appendix C records every arm’s processes, parameters, replication counts, and seeding.

A primary sweep varies the first-stage strength under oracle, estimated-linear, and **estimated-flexible (random-forest)** instrument residualizations, the last reproducing

the flexible- $\hat{\pi}$ regime of the empirical cells; an adversarial stress panel probes six further corners; and an over-identified layer matches the Angrist–Krueger cell’s instrument counts and strengths (§4.9) at $d_z = 3$ and $d_z = 30$, adding the score and conditional-QLR coverage and the J_{CUE} size that case requires. Nonlinear-nuisance, clustered, and survey-weighted arms (weighting moves the estimand there, as in OHIE) hold nominal coverage exactly where naive heteroskedasticity-only and unweighted variants fail badly (Table B.2 prints every coverage value; designs in Online Appendix D.1). A seventh, combined arm runs every ingredient at once: clustered data, the fold-nested constructed instrument of the examiner cell, and nested selection, with an observed-instrument control row (Table B.2). Its one off-nominal corner is the weak-strength constructed-instrument point (0.912, about three Monte-Carlo standard errors below nominal, against 0.956 for the observed-instrument control at the same design strength), the boundary the examiner configuration inherits (§4.5; Table 2 points to this arm). Two observed-instrument controls separate what the construction costs, and neither the clustered design nor the nested selection is implicated. At the same *design* strength the control covers at 0.956, and what the construction costs there is identification strength, cutting the weak point’s median effective- F from 9.6 to 2.0. Lowering the design strength until an observed instrument attains that same effective- F leaves the control at 0.936, so the strength loss and the residual construction error account for about half of the shortfall each, each around two Monte-Carlo standard errors. Where strength survives, the constructed instrument covers: the moderate-strength constructed row is at 0.944 against 0.952 for its observed control, and enlarging the construction support from $m = 6$ to $m = 20$ cases per cluster, relaxing $J_n/\sqrt{n}$ from about 0.79 to 0.43 (Online Appendix C), leaves it at 0.944, as the $J_n = o(\sqrt{n})$ rate condition of Proposition 5 predicts: the corner is a construction-support and strength boundary, not a failure of the cluster-robust AR.

Figure S.1 in Online Appendix D.1 shows the pattern (the strength-sweep panel is Table B.2). Across the oracle, estimated-linear, nonlinear, and over-identified regimes the cross-fitted AR coverage ranged from 0.938 to 0.968 (target 0.95), and across the eleven adversarial

stress points from 0.920 to 0.961 (the per-point panel is Table B.7 in Online Appendix B). In the regime that most resembles the applications (an estimated flexible random-forest $\hat{\pi}$ on a nonlinear design), AR coverage was 0.951 and 0.949 at the two calibrated design points (Online Appendix D.1), with the Wald interval over-covering (0.967 and 0.959) while mislocating the point estimate. The deepest stress dip is the flexible-RF point at effective- $F \approx 33.5$, where AR coverage falls to 0.920 (600 replications, Monte-Carlo standard error 0.011), two to three Monte-Carlo standard errors below nominal; we report it explicitly rather than excluding it, as finite-sample noise from the flexible, generated instrument in the moment and its variance at the null. Wald, by contrast, failed where the theory predicts. In the over-identified design (with oracle, fixed-linear π nuisances; i.e., the classical over-identified weak-IV Wald bias, not a cross-fitting artifact) its coverage fell to 0.91 at $d_z = 5$ and to 0.80 at $d_z = 10$; the identification-robust sets hold their nominal levels at the over-identified operating points, and no Wald values are stored for the $d_z = 3$ and $d_z = 30$ arms (Table B.2). In the just-identified design the Wald point estimate was wildly unstable at weak instruments while an inflated Wald standard error mechanically covered without conveying information (instability figures and the Angrist–Krueger-regime detail: Online Appendix D.1; the real-data counterpart is §4.9). We therefore treat the AR set as the primary interval throughout §4 and report Wald only for contrast.

4.3 Diagnostics and an identification-aware selector

Two further outputs accompany every estimate. First, an **instrument-absorption panel** reports, per learner, R_{oof}^2 of each nuisance, the in-sample-minus-out-of-fold overfit gap for $\hat{\pi}$, the residualized-variance ratio $\text{Var}(\tilde{Z})/\text{Var}(Z)$, and a π -calibration slope, the quantities that separate the routes of §2.4. Second, we implement an **identification-aware π -selector** following the identification-aware cross-validation (IACV) of Çetin (2026b): prediction-optimal residualization can leave *realized* residualized strength on the table at finite n . Within each outer training set, learners within one estimated standard error of the best

out-of-fold predictive risk on a disjoint selection subsample from the tie set, and the selector maximizes a residualized-strength proxy on that subsample alone. The screen keeps the selected learner inside the library’s rate discipline, and the selection statistics never touch the inference fold, so the selector’s *validity* is imported from the IACV results of Çetin (2026b) through the theorems of §3.2; its finite-sample *benefit* is an empirical question we measure rather than assert (Online Appendix E.1).

A paired known-truth benchmark (Figure S.4 in Online Appendix E.1; Table B.3; 300 paired replications per regime, **correctly specified nuisances**, common random numbers) measures what the tie-break changes, and the measured accuracy effects are within Monte-Carlo uncertainty. In the weakest regime (median effective- $F \approx 4$) the point estimates favor the identification-aware rule (root-mean-square error $1.93 \rightarrow 1.52$), but every paired difference is inside its own standard error, and at moderate strength and in the low-absorption do-no-harm regime the two rules are equivalent to the resolution of the design; Table B.3 prints the paired differences with their standard errors regime by regime. Anderson–Rubin coverage lies between 0.93 and 0.97 for both rules in every regime, with rule differences within Monte-Carlo standard error; the tie-break does change selections (Online Appendix E.1 reports the winner shares). We accordingly present the selector as an **exploratory refinement**: its validity is inherited from the leakage-safe protocol and the IACV results cited above, while a finite-sample accuracy benefit is suggested by the weak-regime point patterns but is not demonstrated at this simulation size, and we make no dominance claim.

Because $\hat{\pi}$ is chosen for strength on data also used, downstream, to form the Anderson–Rubin set, the selection could in principle distort the reported coverage. By construction it should not: under IACV the selected $\hat{\pi}$ is measurable with respect to data disjoint from the inference fold, so the null behavior of the AR statistic is unchanged by the selection (Çetin 2026b, Lemma 2). Empirically, a matched split-sample experiment compares the AR coverage of τ_0 under in-sample and disjoint-half selection of the π -learner: across the three regimes the selection distortion is at most 0.016 in absolute value, the same size for the pure predictive

rule as for the identification-aware one (both 0.016 in the moderate regime), and comparable to the Monte-Carlo standard error of the 120-replication coverage estimates (about 0.02): the check reads as sampling noise, not a selection effect. One design caveat applies: this contrast varies selection reuse and winner aggregation together. A foldwise pure control that varies only the selection-inference overlap gives distortions from -0.008 to $+0.017$ across the three regimes and both rules, again within the roughly 0.02 Monte-Carlo standard error, so the small distortions are not attributable to the winner-aggregation difference. Online Appendix E.2 records both designs and the residual departure both modes inherit from the generated-instrument effect of §4.2.

4.4 OHIE: a randomized negative control

The Oregon Health Insurance Experiment (Finkelstein et al. 2012) assigned Medicaid eligibility by lottery, so the instrument Z (lottery selection) is randomized conditional on household size. By §2.4, $\pi_0(W)$ is constant within the conditioning strata of the lottery population, closing the population absorption channel; in the responder sample below this constancy is an approximation the measured $R_{\text{oof}}^2(\hat{\pi}; Z) \approx 0$ supports. Finite-sample overfit dilution remains possible in principle, and the diagnostics below show it is not materially present. D is ever-enrollment in Medicaid, Y a twelve-month survey outcome (primary-care visits), W includes the required household-size dummies, and clustering is on household. Following Finkelstein et al. (2012), we take the **weighted-responder sample as the primary specification**: the 23,441 twelve-month survey responders with complete analysis variables, weighted by their survey weights; the weighted moment targets τ_w , the PLIV coefficient under the survey-weighted responder law (Online Appendix D.2). By construction the weights carry survey-wave and lottery-draw components beyond W , so they are not a function of the controls alone (a linear regression of the weight on W attains an R^2 of only 0.03, consistent with this), and the weighted target law is carried through the whole pipeline, not only the final moment. The primary specification thus implements regime (b) of Proposition 4; under

its hypotheses the survey-weighted, household-clustered AR set inherits the uniform validity of Theorem 2 at any tested value satisfying the weighted null W-ID(t).

In the weighted-responder primary specification the residualized effective- F was approximately 2100 and stayed flat across the learner library (Table D.2; Figure S.2 in Online Appendix D.2), and the estimate was stable at $\hat{\tau} \approx 1.04$ additional primary-care visits (ordinary least squares 1.20, two-stage least squares 1.03; unweighted, a slightly lower 0.97). The absorption diagnostics agree: every learner, including the random forest, had $R_{\text{of}}^2(\hat{\pi}; Z) \approx 0$ and $\text{Var}(\tilde{Z})/\text{Var}(Z) \approx 1$, so there was nothing to absorb and no material overfit dilution. A collapse here would have indicated a finite-sample DML-IV pathology; its absence shows the pipeline does not manufacture one under randomization and strong identification; the sources of the erosions in the other cells are traced by their own diagnostics (§4.5–§4.8).

The lottery raises survey response in the raw sample, so the survey sample is itself selected on treatment; under the survey weights the response shares are nearly equal (Online Appendix D.2 reports the shares). Lee (2009) trimming bounds are therefore computed in the weighted frame. They trim the slightly over-responding control arm by the implied 2.2% and place the instrumented effect on visits within $[0.77, 2.05]$ (Table B.5), an interval that brackets the trimmed point estimate of 0.92 but has a wide upper bound: under the weighted worst case the trimmed interval for the *outcome* effect lies above zero, a bound on the point-identified object, not a confidence statement, and far from tight. In the unweighted panel the interval $[-4.0, 4.3]$ contains zero and is uninformative (Table B.5). This sensitivity attaches to the survey outcome, not to the cell’s role as a negative control for residualized concentration: the effective- F stays in the thousands and flat across learners whatever the outcome’s selection, the only property the mechanism argument relies on.

4.5 Examiner leniency: overfit dilution

The examiner cell uses Miami–Dade pretrial data from the Chyn, Frandsen, and Leslie (2025) replication package, restricted, following that study, to judges with at least 200 cases

(91,282 of 94,210), with Y a conviction indicator, D pretrial release, and a leave-one-shift-out (cluster-jackknife) judge-leniency instrument, clustering on the court-by-date assignment (the bond-hearing shift) and absorbing court-by-time fixed effects into W . The leniency instrument is itself a generated, cross-fitted object, entering the protocol as a fourth trained nuisance (Online Appendix D.3; its finite-sample estimation noise is what the combined seventh arm of §4.2, Table B.2, evaluates). This fold-nested construction is a hypothesis of Proposition 5, under which the cluster-robust AR set retains the uniform validity of Theorem 2; its further conditions are the exclusion restriction G-EX and a leniency-estimation rate asking the judge dimension to grow slowly ($J_n = o(\sqrt{n})$), with no lower bound imposed on the generated instrument’s relevance. Because pretrial release is the sign-flipped complement of detention, a negative $\hat{\tau}$ means detention raises conviction. Here the instrument is generated at the judge level while W carries court-by-time fixed effects and case covariates, so $\hat{\tau}$ chases judge-level variation only through those proxies; overfit dilution of the instrument is an economically real danger; machine-learned instruments in this design class are known to carry finite-sample hazards (Angrist and Frandsen 2022).

We validated the full-sample construction against the published Stata implementation: our cluster-jackknife first-stage coefficient was 0.319 against their 0.314, and our just-identified estimate was about -0.43 against their reported -0.432 ; the decoupled-nuisance pipeline of Table 1 gives a DML-ridge $\hat{\tau} \approx -0.48$. The full-sample construction thus reproduces an independent reference implementation; the fold-nested instrument is exercised separately by its test-suite fold-nesting contracts and the known-truth combined arm of §4.2 (Table B.2).

The flexible $\hat{\pi}$ then revealed the **dilution** channel rather than absorption. The random-forest $\hat{\pi}$ could not predict the orthogonalized leniency instrument out of fold ($R_{\text{oof}}^2(\hat{\pi}; Z) < 0$) though it fit nearly half its variance in sample (Table B.12), and cross-fitting turned that overfit into noise: $\text{Var}(\tilde{Z})/\text{Var}(Z) = 1.26$, above one, and the residualized effective- F fell from 20.1 under a ridge π to 14.2 under the random forest (Figure S.3 in Online Appendix D.3). The estimate itself moved little across the learner library (-0.44 to -0.48 , within one Wald

standard error), but the inference did not: under the random forest the cluster-robust standard error grows from 0.22 to 0.26 and the Anderson–Rubin set widens from $[-1.02, -0.04]$ to $[-1.12, 0.10]$, no longer excluding zero, flagging the flexible $\hat{\pi}$ as the unreliable choice for this cell. Resampling the cross-fitting partition moves the estimate more than the learner library does and leaves the robust set containing zero in two of six partitions (Table B.12), so the exclusion of zero here is a property of the reported partition, not of the design alone.

4.6 CollegeDistance: a leakage-corrected cautionary cell

The CollegeDistance cell (Rouse 1995) uses distance to the nearest college as the instrument; it is the application in which Çetin (2026a) documents the collapse. Under ordinary shuffled cross-validation a flexible $\hat{\pi}$ appears to absorb the instrument, with out-of-fold $R_{\text{oof}}^2(\hat{\pi}; Z) = 0.78$; with cluster-aware folds on the 43-group state proxy the diagnosis reverses, R_{oof}^2 falling to -0.07 with a large in-sample overfit gap (0.94): the apparent absorption was geographic fold leakage, and with the leakage removed the cell is simply weak. The cell is a stress test of the cross-fitting protocol rather than a clean mechanism cell: the outcome is a state-level index no outcome learner predicts out of fold, and the coupled-learner disagreement came from the nuisances, not the instrument, disappearing once each nuisance is fit by its own MSE-best learner. We therefore report CollegeDistance as a leakage-and-misspecification illustration that motivates cluster-aware cross-fitting and separate nuisance learners. Online Appendix D.4 records the complete cell.

4.7 Card: a clean schooling cell, and a within-instrument concentration ladder

The Card (1995) cell replaces CollegeDistance’s state-level outcome with an individual-level one: Y is log hourly wage, D is years of education, and Z indicates growing up near a four-year college (`nearc4`), so the estimand is a genuine return to schooling. With the standard

controls, on the full sample of 3,010 men, the just-identified two-stage least-squares estimate is $\hat{\tau} = 0.13$ (SE 0.05), the canonical Card magnitude. The cell behaves as Proposition 1 predicts: the decoupled-nuisance estimate is **stable across the π library** (0.128 linear, 0.152 random forest, 0.118 gradient boosting), with the residualized effective- F near 15–18 and bounded, overlapping Anderson–Rubin sets (about $[0.03, 0.27]$ for the ridge baseline). Here `nearc4` retains residualized strength, so a flexible $\hat{\pi}$ neither absorbs nor dilutes it.

Card also affords a controlled view of how residualized concentration is *built*: the same instrument can be made strong or weak by the conditioning set. Holding the instrument fixed (`nearc4`, $d_z = 1$) and the sample fixed (the 2,182 men with the full background battery observed), we climb a five-rung control-richness ladder from lean controls to a degree-two expansion (Online Appendix D.5 lists the rungs). The upper rungs mix predetermined and 1976-measured controls (experience is age minus schooling minus six), so the ladder is read as concentration formation, not competing causal specifications; replacing experience with age moves the canonical estimate by about a third and carries its robust set across zero (Table B.11). The residualized effective- F falls from about 7.5 to about 2.5 (non-monotonically; Table B.4 and Online Appendix D.5), and the Anderson–Rubin set widens from bounded ($[-0.15, 0.27]$) to the whole real line. The decomposition is itself diagnostic: across the ladder the residualized-instrument variance $\text{Var}(\tilde{Z})$ and the $R_{\text{sof}}^2(\hat{\pi}; Z)$ are essentially flat (Online Appendix D.5 prints the ranges), so the loss is *not* instrument absorption; what falls is the first-stage slope $\text{Cov}(\tilde{Z}, \tilde{D})/\text{Var}(\tilde{Z})$, from 0.26 to 0.12. The background controls absorb the Z – D covariance through the *treatment* nuisance $r(W)$, a route to concentration loss distinct from the π -channels of §2.4. The point estimate does not visibly collapse: $\hat{\tau}$ drifts only within approximately $[0.04, 0.08]$ along the entire linear- π ladder (Online Appendix D.5 reports the wider random-forest drift), even as the instrument becomes uninformative, so the identification-robust set registers what it does not. At $d_z = 1$ the §3.2 inversion is analytic: it labels the AR set unbounded exactly when the tail limit of the AR statistic falls below the χ_1^2 critical value 3.84. That is the case at the three richest rungs, where the effective- F has

fallen to 1.85, 3.52, and 2.51, and the Anderson–Rubin set at each is the whole line (Table B.4).

This ladder reconciles our clean cell with the returns-to-schooling literature’s mixed experience of Card. In the richer specification under which Card is most often judged weak in that literature (Online Appendix D.5 gives it), the joint first-stage F is 4.69 and the estimate is an uninformative null (two-stage least squares -0.03 , decoupled DML -0.01), which our pipeline reproduces to the reported digits; every step past the lean baseline leaves the design too weak to inform τ . Online Appendix D.5 traces how the gap between this null and the canonical 0.13 tracks the ladder. Card is therefore not a contradictory data point but the low-concentration end of a single concentration ladder. Which specification to report is itself an identification question with two parts in a fixed order. The instrument’s validity and the estimand each specification targets come first; they are maintained assumptions that no strength statistic can supply. Given them, the concentration object C and the Anderson–Rubin set say how much the data can inform τ under each specification.

4.8 Cross-cell synthesis

Across the real-data cells, flexible nuisance learning leaves the residualized first stage intact wherever the instrument carries genuine residualized strength; the apparent erosions arise from finite-sample estimation (overfit dilution) or from the conditioning set and the treatment-nuisance specification, never, in population, from a correctly specified flexible π . It is flat under randomization (OHIE, §4.4) and for a real but moderately weak schooling instrument once the cross-fitting is disciplined (the Card baseline, §4.7); it erodes through *overfit dilution* where the instrument is itself a high-dimensional estimated object (the examiner cell, §4.5); and it erodes along the Card ladder through the treatment nuisance $r(W)$, so that concentration is a property of the conditioning set, not of the instrument in the abstract (§4.7). The one cell that appeared to show genuine absorption, CollegeDistance, was geographic fold leakage; corrected, it is a weak cell whose flexible-learner movement is an overfitting artifact (§4.6).

No real-data cell instantiates *genuine* absorption.

The known-truth Monte Carlo shows what the apparent “absorption” is: with a designed nonlinear π_0 and known τ_0 , a random forest recovers it out of fold where a linear learner does not, and both estimators are essentially unbiased (Online Appendix D.1 prints the fit and bias figures and which formal result covers each arm). The effective- F , however, turns on how the *treatment* nuisance is specified. Under a correctly specified $r = \mathbb{E}[D \mid W]$ the flexible $\hat{\pi}$ attains the **higher** effective- F (about 57 against 34 for a ridge π), the sample counterpart of Proposition 3’s population ranking of C , while the “absorption” ordering, in which the flexible $\hat{\pi}$ shows the *lower* effective- F (about 93 against 152), appears only when $\hat{r}$ is fit by a linear learner that underfits the nonlinear $\mathbb{E}[D \mid W] = r_0 + f_s\pi_0$ (with f_s the design’s first-stage loading): the unmodeled $f_s\pi_0$ structure remains in $\tilde{D}$ and is correlated with the W -predictable instrument variation a flexible π removes. The drop is therefore a treatment-nuisance specification artifact, not a population cost of flexible π , removed by specifying r flexibly. It does not fade with the sample size alone: a fixed misspecified linear $\hat{r}$ leaves the unmodeled $f_s\pi_0$ structure in $\tilde{D}$ at every n , and the discrepancy shrinks only along sequences in which the r -learner is consistent for $\mathbb{E}[D \mid W]$ or the misspecification itself vanishes. The treatment-nuisance specification is thus the first diagnostic to check.

The synthesis is Figure 1, and its caption states the pattern; the sign of R_{of}^2 and the residualized-variance ratio separate the routes. In the CollegeDistance cell the cluster-aware fold discipline changes the *diagnosis*, not the point estimate: the apparent genuine-absorption collapse is revealed as geographic fold leakage while the flexible- $\hat{\pi}$ point estimate stays near zero, so the correction relocates the cause rather than rescuing the estimate. Decoupling the nuisances is a separate margin that does move estimates: it removed the coupled-learner disagreement in that cell (§4.6). The practical lesson is that a near-zero estimate under flexible learning is a statement about residualized concentration and inference, not a defect of machine learning per se. Such an estimate is to be read with the absorption diagnostics, an identification-robust interval, and disciplined cross-fitting.

4.9 Angrist–Krueger: over-identified weak instruments

The four cells above are just-identified ($d_z = 1$), where the Anderson–Rubin, score, and conditional-QLR sets coincide (§3.2); the Angrist and Krueger (1991) quarter-of-birth design supplies the over-identified counterpart. The cell tests the over-identified weak-instrument inference of §3.2: with the three quarter-of-birth dummies ($d_z = 3$) the design is informative, yielding the canonical 0.108 with bounded, mutually close AR, score, and conditional-QLR sets (Table B.6). With the thirty quarter-of-birth by year-of-birth interactions ($d_z = 30$), the many-weak specification of Bound, Jaeger, and Baker (1995), two-stage least squares moves toward ordinary least squares with spurious Wald precision, while the identification-robust sets report the design as uninformative about τ , and structurally so: a clustered AR statistic’s analytic tail limit is bounded above by the cluster count, and at $G = 51$ with thirty instruments the χ^2 critical value sits so close to that ceiling that no bounded set is attainable at any strength (Online Appendix C.5). The clustered theorems reach the cell through the within-transformation bridge of Lemma 9 (Online Appendix A.4) and its W conditions on the state bundles. The two theorems are $G \rightarrow \infty$ statements; with $G = 51$ state-of-birth clusters the cell sits at the boundary of that regime, so we do not read them as finite- G evidence here, and report instead a Monte-Carlo arm matched to this cluster structure, in which the three sets hold their nominal levels at $d_z = 3$ (Table B.10). Online Appendix D.6 records the complete cell.

5. Discussion

Our results recast a documented fragility of DML-IV as a problem of inference and design, not identification. When flexible instrument residualization drives a PLIV estimate toward zero with a small standard error, the evidence here suggests that the right reading is not a precise zero but a loss of residualized concentration, which invalidates the Wald interval while leaving the estimand unchanged.

Correctly specified, flexible learning carries no population concentration cost (Proposition 3). Overfit dilution is an estimator pathology that a restricted or more regularized $\hat{\pi}$ removes (Proposition 1; Online Appendix A.1). A richer conditioning set instead lowers conditional relevance (a modeling choice), and an under-specified $\hat{r}$ can manufacture an apparent absorption. The diagnostic signatures of §2.4 separate the cases: regularize $\hat{\pi}$, respecify $\hat{r}$, or accept a weaker conditional design.

Proposition 3 is related to, but distinct from, classical optimal-instruments theory (Amemiya 1974; Chamberlain 1987; Newey 1990). That theory chooses the instrument *transformation* $h(Z, W)$ to attain the efficiency bound under strong identification; here the instrument is held fixed and the choice object is the *residualization* π . Within the residualization class the two aims coincide under conditional homoskedasticity of the score, where $\text{Var}(\hat{\tau}) \propto 1/(nC)$ is exact (Online Appendix A.1); under bounded conditional heteroskedasticity C ranks weak-identification strength up to constants rather than tracking the sandwich variance, which is the sense in which Proposition 3 is a weak-identification result rather than an efficiency result. We develop this delineation formally in follow-up work.

The conclusions are robust along four dimensions: the nuisance-learner library, the control and instrument sets, the cross-fitting protocol under stress, and nonlinear simulation designs, with the checks reported within the cells: learner stability and its exception (§4.5, §4.7, Table 1), control- and instrument-set sensitivity (§4.7, §4.9), fold and decoupling checks (§4.6), and nonlinear-design coverage (§4.2).

Several limitations bound these conclusions. While the concentration result is a theorem (Proposition 3), the *attribution* of a realized loss to overfit dilution versus treatment-nuisance misspecification is an operational characterization supported by diagnostics, and the finite-sample boundary between a correctly recovered nonlinear π_0 and overfitting is not sharp. The estimand is the homogeneous-coefficient PLIV parameter throughout, and that specification is an approximation: under effect heterogeneity a binary-treatment cell (OHIE, examiner) would carry a complier-average reading only under an additional monotonicity condition we

do not impose, and a multivalued-treatment cell (Card, CollegeDistance, Angrist–Krueger) an average-causal-response-type weighted average, as Table 2 records. The Monte-Carlo coverage evaluation is calibrated to the regimes the cells occupy (effective- F from about one to about five hundred in the simulation sweep, instrument dimension up to thirty, the over-identified Angrist–Krueger regimes included) and does not claim universal validity; in particular, a second-order product of nuisance estimation errors enters the cross-fitted Anderson–Rubin moment with a factor of the tested τ , so the τ -scaled product term is largest, and the product-rate condition most binding, at the far endpoints of the set. Finally, the CollegeDistance cell carries a region-determined outcome and therefore serves as a leakage-and-misspecification stress test rather than a clean estimand; the clean schooling estimand is supplied by the Card cell (§4.7).

A locally robust corrected Anderson–Rubin moment (Chernozhukov, Escanciano, Ichimura, Newey, and Robins 2022) would suppress the second-order nuisance products at every tested value, rather than controlling them through the product-rate condition of Theorems 1 and 2 (Online Appendix A.4). A formal dominance result for the identification-aware selector, beyond the Monte-Carlo evidence, would complete the constructive analysis.

6. Conclusion

We have argued that the collapse of a DML-IV estimate under flexible nuisance learning is a loss of residualized concentration, not of point identification of the estimand. With a correctly specified treatment nuisance it is not an inherent cost of flexible learning: prediction-optimal residualization maximizes the population concentration (Proposition 3). The realized losses run through sources that standard diagnostics tell apart: overfit dilution, the conditioning set, and treatment-nuisance specification. We executed the weak-instrument-robust inference this requires. The cross-fitted Anderson–Rubin set retains uniform asymptotic size under clustering (Theorem 2, Online Appendix A.4), under survey weights, and with the fold-nested

generated instrument (Propositions 4 and 5, Online Appendix A.6). The i.i.d. baseline is imported from Çetin (2026b). The score statistic carries the clustered and weighted guarantees under an added non-degeneracy condition (Theorem 3, Online Appendix A.5), and the generated-instrument one only at $d_z = 1$, where it coincides with Anderson–Rubin. A known-truth Monte Carlo showed the cross-fitted Anderson–Rubin set keeps coverage close to nominal across the regimes our applications occupy and, apart from two stress corners reported in §4.2, closer to nominal than Wald where identification is weak. There Wald under-covers (over-identified) or inflates the standard error of an unstable estimate (just-identified). A structured diagnostic portfolio, from a randomized lottery through a high-dimensional examiner instrument to a weak proximity instrument, located overfit dilution in the examiner data and the conditioning-set route in the Card ladder; in the randomized control neither is materially present, and the simulation showed the apparent absorption to be a treatment-nuisance artifact. An exploratory identification-aware π -selector broke predictive-loss ties toward the stronger selection-fold proxy without disturbing the robust inference; its measured accuracy effects stayed within simulation uncertainty.

The takeaway is the reading of §5: absent a strong residualized first stage, a near-zero estimate with a small standard error under flexible nuisance learning is a weak-instrument signal rather than a precise zero, reported with an identification-robust interval and absorption diagnostics locating the source. The concentration result is a theorem (Proposition 3); the attribution of a realized loss to its source is diagnostic. The distributional guarantees of Online Appendix A.4–A.6 are asymptotic statements under their assumption packages, and the simulated coverage is calibrated to the regimes the data occupy, not universal. The over-identified Angrist–Krueger cell (§4.9) is a real-data counterpart to the simulated many-weak failure: two-stage least squares moves toward ordinary least squares, as Bound, Jaeger, and Baker (1995) predict, while the identification-robust sets report the design as uninformative. Within those bounds, the analysis turns a documented fragility of machine-learning IV estimation into a measurable, partly correctable property of the residualized first stage.

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Data and code availability

The replication materials reproduce every number, table, and figure in the paper from the result artifacts included with the package. Every Monte-Carlo row derives from a single master seed through per-row keyed streams and, in the locked environment recorded in `REPRODUCE.md`, reproduces to machine-level floating point, with BLAS-threading differences confined below every reported digit; the deterministic cells (Card, CollegeDistance, Angrist–Krueger) reproduce exactly in that environment, and the examiner cell’s random-forest fold sampling is sensitive to the scikit-learn build, which can move its effective- F by about 0.1 across builds. Three public datasets are included with the code (`CollegeDistance`, Card, Angrist–Krueger); two are governed by their providers’ access terms and must be obtained from source (the Oregon Health Insurance Experiment from NBER and the Miami–Dade examiner data from openICPSR 209883), with placement instructions in the accompanying `REPRODUCE.md`. The residualized-strength, Monte-Carlo coverage, selector, and over-identified analyses are produced by the scripts in `Reproduction code/` and orchestrated by `run_all.py` (whose restricted mode stops with an explicit error on any missing input); all figures are regenerated from the included JSON artifacts by `make_figures.py`.

References

- Amemiya, T. (1974). The nonlinear two-stage least-squares estimator. *Journal of Econometrics* 2(2): 105–110. doi:10.1016/0304-4076(74)90033-5
- Anderson, T. W., and H. Rubin (1949). Estimation of the parameters of a single equation in a complete system of stochastic equations. *Annals of Mathematical Statistics* 20(1): 46–63. doi:10.1214/aoms/1177730090
- Andrews, I., J. H. Stock, and L. Sun (2019). Weak instruments in instrumental variables regression: Theory and practice. *Annual Review of Economics* 11: 727–753. doi:10.1146/annurev-economics-080218-025643
- Angrist, J. D., and B. Frandsen (2022). Machine labor. *Journal of Labor Economics* 40(S1): S97–S140. doi:10.1086/717933
- Angrist, J. D., and A. B. Krueger (1991). Does compulsory school attendance affect schooling and earnings? *Quarterly Journal of Economics* 106(4): 979–1014. doi:10.2307/2937954
- Athey, S., J. Tibshirani, and S. Wager (2019). Generalized random forests. *Annals of Statistics* 47(2): 1148–1178. doi:10.1214/18-AOS1709
- Avila Márquez, M. (2026). Weak instrumental variables due to ignored nonlinearities in panel data: A Super Learner Control Function estimator. Working paper, arXiv:2504.03228 (first posted April 2025, revised July 2026). doi:10.48550/arXiv.2504.03228
- Baiardi, A., P. S. Clarke, A. A. Naghi, and A. Polselli (2026). Double machine learning for static panel data with instrumental variables: New method and applications. Working paper, arXiv:2603.20464. doi:10.48550/arXiv.2603.20464
- Belloni, A., D. Chen, V. Chernozhukov, and C. Hansen (2012). Sparse models and methods for optimal instruments with an application to eminent domain. *Econometrica* 80(6): 2369–2429. doi:10.3982/ECTA9626

- Blandhol, C., J. Bonney, M. Mogstad, and A. Torgovitsky (2026). When is TSLS actually LATE? *Review of Economic Studies*, forthcoming. doi:10.1093/restud/rdag029
- Bound, J., D. A. Jaeger, and R. M. Baker (1995). Problems with instrumental variables estimation when the correlation between the instruments and the endogenous explanatory variable is weak. *Journal of the American Statistical Association* 90(430): 443–450. doi:10.1080/01621459.1995.10476536
- Card, D. (1995). Using geographic variation in college proximity to estimate the return to schooling. In L. N. Christofides, E. K. Grant, and R. Swidinsky (eds.), *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*, 201–222. University of Toronto Press.
- Çetin, T. (2026a). Nested Cross-Validated Double Machine Learning for the Partially Linear IV Model. Manuscript submitted to *The Econometrics Journal*. Posted on ResearchGate, doi:10.13140/RG.2.2.18811.22562
- Çetin, T. (2026b). Adaptive Nuisance Selection for Weak-Identification-Robust Inference. Working paper, University of California, Berkeley. https://tamercetin.com/Adaptive_Nuisance_Selection_for_Weak-Identification-Robust_Inference_Combined.pdf
- Çetin, T. (2026c). Feasible Power Frontiers for Learned Scalar Anderson–Rubin Scores under High-Dimensional Weak Identification. Manuscript submitted to *Econometrica*. https://tamercetin.com/Learned_Scalar_AR_Power_Frontiers_Combined.pdf
- Chamberlain, G. (1987). Asymptotic efficiency in estimation with conditional moment restrictions. *Journal of Econometrics* 34(3): 305–334. doi:10.1016/0304-4076(87)90015-7
- Chao, J. C., and N. R. Swanson (2005). Consistent estimation with a large number of weak instruments. *Econometrica* 73(5): 1673–1692. doi:10.1111/j.1468-0262.2005.00632.x
- Chernozhukov, V., D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrics Journal* 21(1): C1–C68. doi:10.1111/ectj.12097

- Chernozhukov, V., J. C. Escanciano, H. Ichimura, W. K. Newey, and J. M. Robins (2022). Locally robust semiparametric estimation. *Econometrica* 90(4): 1501–1535. doi:10.3982/ECTA16294
- Chyn, E., B. Frandsen, and E. C. Leslie (2025). Examiner and judge designs in economics: A practitioner’s guide. *Journal of Economic Literature* 63(2): 401–439. doi:10.1257/jel.20241719
- Davidson, R., and J. G. MacKinnon (2014). Confidence sets based on inverting Anderson–Rubin tests. *Econometrics Journal* 17(2): S39–S58. doi:10.1111/ectj.12015
- Dobbie, W., J. Goldin, and C. S. Yang (2018). The effects of pretrial detention on conviction, future crime, and employment: Evidence from randomly assigned judges. *American Economic Review* 108(2): 201–240. doi:10.1257/aer.20161503
- Finkelstein, A., S. Taubman, B. Wright, M. Bernstein, J. Gruber, J. P. Newhouse, H. Allen, K. Baicker, and the Oregon Health Study Group (2012). The Oregon health insurance experiment: Evidence from the first year. *Quarterly Journal of Economics* 127(3): 1057–1106. doi:10.1093/qje/qjs020
- Hansen, C., J. Hausman, and W. Newey (2008). Estimation with many instrumental variables. *Journal of Business & Economic Statistics* 26(4): 398–422. doi:10.1198/073500108000000024
- Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica* 50(4): 1029–1054. doi:10.2307/1912775
- Hartford, J., G. Lewis, K. Leyton-Brown, and M. Taddy (2017). Deep IV: A flexible approach for counterfactual prediction. *Proceedings of ICML*, 1414–1423.
- Kleibergen, F. (2005). Testing parameters in GMM without assuming that they are identified. *Econometrica* 73(4): 1103–1123. doi:10.1111/j.1468-0262.2005.00610.x
- Lee, D. S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *Review of Economic Studies* 76(3): 1071–1102. doi:10.1111/j.1467-

- Lee, D. S., J. McCrary, M. J. Moreira, and J. Porter (2022). Valid t -ratio inference for IV. *American Economic Review* 112(10): 3260–3290. doi:10.1257/aer.20211063
- Lim, D., W. Wang, and Y. Zhang (2024). A dimension-agnostic bootstrap Anderson–Rubin test for instrumental variable regressions. Working paper, arXiv:2412.01603. doi:10.48550/arXiv.2412.01603
- Ma, Y. (2026). Identification-robust inference for the LATE with high-dimensional covariates. *Journal of Econometrics* 257: 106302. doi:10.1016/j.jeconom.2026.106302
- Mikusheva, A., and L. Sun (2022). Inference with many weak instruments. *Review of Economic Studies* 89(5): 2663–2686. doi:10.1093/restud/rdab097
- Montiel Olea, J. L., and C. Pflueger (2013). A robust test for weak instruments. *Journal of Business & Economic Statistics* 31(3): 358–369. doi:10.1080/00401706.2013.806694
- Moreira, M. J. (2003). A conditional likelihood ratio test for structural models. *Econometrica* 71(4): 1027–1048. doi:10.1111/1468-0262.00438
- Navjeevan, M. (2025). An identification- and dimensionality-robust test for instrumental variables models. Working paper, Texas A&M University (arXiv:2311.14892; revised June 24, 2025). doi:10.48550/arXiv.2311.14892
- Newey, W. K. (1990). Efficient instrumental variables estimation of nonlinear models. *Econometrica* 58(4): 809–837. doi:10.2307/2938351
- Rouse, C. E. (1995). Democratization or diversion? The effect of community colleges on educational attainment. *Journal of Business & Economic Statistics* 13(2): 217–224. doi:10.1080/07350015.1995.10524596
- Sanderson, E., and F. Windmeijer (2016). A weak instrument F -test in linear IV models with multiple endogenous variables. *Journal of Econometrics* 190(2): 212–221. doi:10.1016/j.jeconom.2015.06.004

- Sargan, J. D. (1958). The estimation of economic relationships using instrumental variables. *Econometrica* 26(3): 393–415. doi:10.2307/1907619
- Staiger, D., and J. H. Stock (1997). Instrumental variables regression with weak instruments. *Econometrica* 65(3): 557–586. doi:10.2307/2171753
- Stock, J. H., and J. H. Wright (2000). GMM with weak identification. *Econometrica* 68(5): 1055–1096. doi:10.1111/1468-0262.00151
- Wang, W., and Y. Zhang (2024). Wild bootstrap inference for instrumental variables regressions with weak and few clusters. *Journal of Econometrics* 241(1): 105727. doi:10.1016/j.jeconom.2024.105727